

MANISH KUMAR YADAV

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EDUCATION

Pulchowk Campus, IOE, Tribhuvan University

Bachelor's Degree in Electronics, Communication and Information Engineering
Grade: 79.5%

Lalitpur, Kathmandu

Apr 2021 - Apr 2025

PROFESSIONAL EXPERIENCE

Telemed Nepal

[Web](#)

Full Stack Developer: TypeScript, React, PostgreSQL, Supabase

Feb 2026 - Present

- Built a three-role app (patient, doctor, admin) with Google OAuth and email auth backed by Supabase RLS
- Implemented a multi-state consultation workflow with Supabase Realtime subscriptions for live status updates across all dashboards
- Handled file uploads for prescriptions and payment receipts
- Served 1200+ visits and 28 consultations in first 2 months

Himalaya College of Engineering, Tribhuvan University

Kathmandu

Assistant Lecturer

May 2025 - Apr 2026

- Instructed over 200+ undergraduate students in courses on Data Communication, Digital Logic, Filter Design, Embedded Systems, Digital Logic, Microprocessor
- Evaluated quizzes, assignments, laboratory works and assessments

HexCode Nepal

Kathmandu

ML Engineer Intern: Python, Scikit-learn

Feb 2023 - Aug 2023

- Implemented supervised, unsupervised, and reinforcement learning algorithms in Python
- Developed spam filter algorithm for text messages using binary classification
- Developed Kathmandu land price prediction algorithm using linear regression

OPENSOURCE CONTRIBUTION

ns-3-dev: A discrete event network simulator (540 stars)

[GitLab](#)

Contribution: Fixed outdated header path references in ns-3's documentation (C++)

Apr 2026

PROJECTS

shellsense: a CLI tool to explain Linux shell commands in plain English

[GitHub](#)

Python

Mar 2026

- Built a command parser that extracts commands, subcommands, flags and arguments. Also handles pipelines and redirects.
- Implemented multi-layer resolution: knowledge base (a python dictionary), cache, man pages, tldr
- Added a caching layer using JSON files with 30-day TTL to avoid repeated man page parsing and network calls
- Rendered terminal output using the Rich library

job-aggregator: a job listings scraper

[GitHub](#)

Python

Apr 2026

- Built a Python automation tool that scrapes job listings from multiple sources, removes duplicates, scores the listings against resume keywords, and emails a daily digest using GitHub Actions
- Designed a keyword-matching scoring algorithm that ranks listings against resume keywords
- Built a URL-based deduplication layer to filter repeat postings across sources
- Automated a daily HTML email digest over SMTP, with HTML escaping to prevent XSS injection
- Scheduled daily execution via GitHub Actions workflow

p2p-transfer: browser based app to send files directly between two devices

[GitHub](#) [Web](#)

JavaScript, Node.js, WebRTC

Dec 2025

- Built a serverless P2P file transfer using WebRTC
- Implemented a Node.js WebSocket signaling server for SDP/ICE exchange
- Chunked streaming (64KB per chunk) with backpressure control for large files
- SHA-256 password-protected rooms, QR-code based link sharing, real-time speed display

RISC single cycle and pipelined processor

 [GitHub](#)  [GitHub](#)

Verilog, GTKWave

Sep 2025

- Implemented program counter, instruction memory, register file, data memory, ALU and control unit
- Implemented five-stage pipelined processor with hazard control

TECHNICAL SKILLS

- **Programming:** Python, C++, C, JavaScript, Assembly, Shell, SQL, Verilog, MATLAB
- **Developer Tools:** VS Code, Git/GitHub, Linux, Docker
- **Frameworks & Libraries:** React, Node.js, WebRTC, Supabase

RESEARCH EXPERIENCE

MDPI JCP

 [Web](#)

10 citations, 6800 reads

Mar 2025

- Machine Learning-Based Attack Detection and Mitigation with Multi-Controller Placement Optimization over SDN Environment

IEEE Access

 [Web](#)

1 citation, 1133 reads

Mar 2025

- Game Theoretic Approach to QoS Oriented Machine Learning Model Development towards 5G Network Migration Planning.

CERTIFICATIONS

- [Coursera: Machine Learning Specialization \(Stanford | Online\)](#)
- [Coursera: Python for Everybody Specialization \(University of Michigan\)](#)